



**WORKHORSE™**

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# First Responder Guide

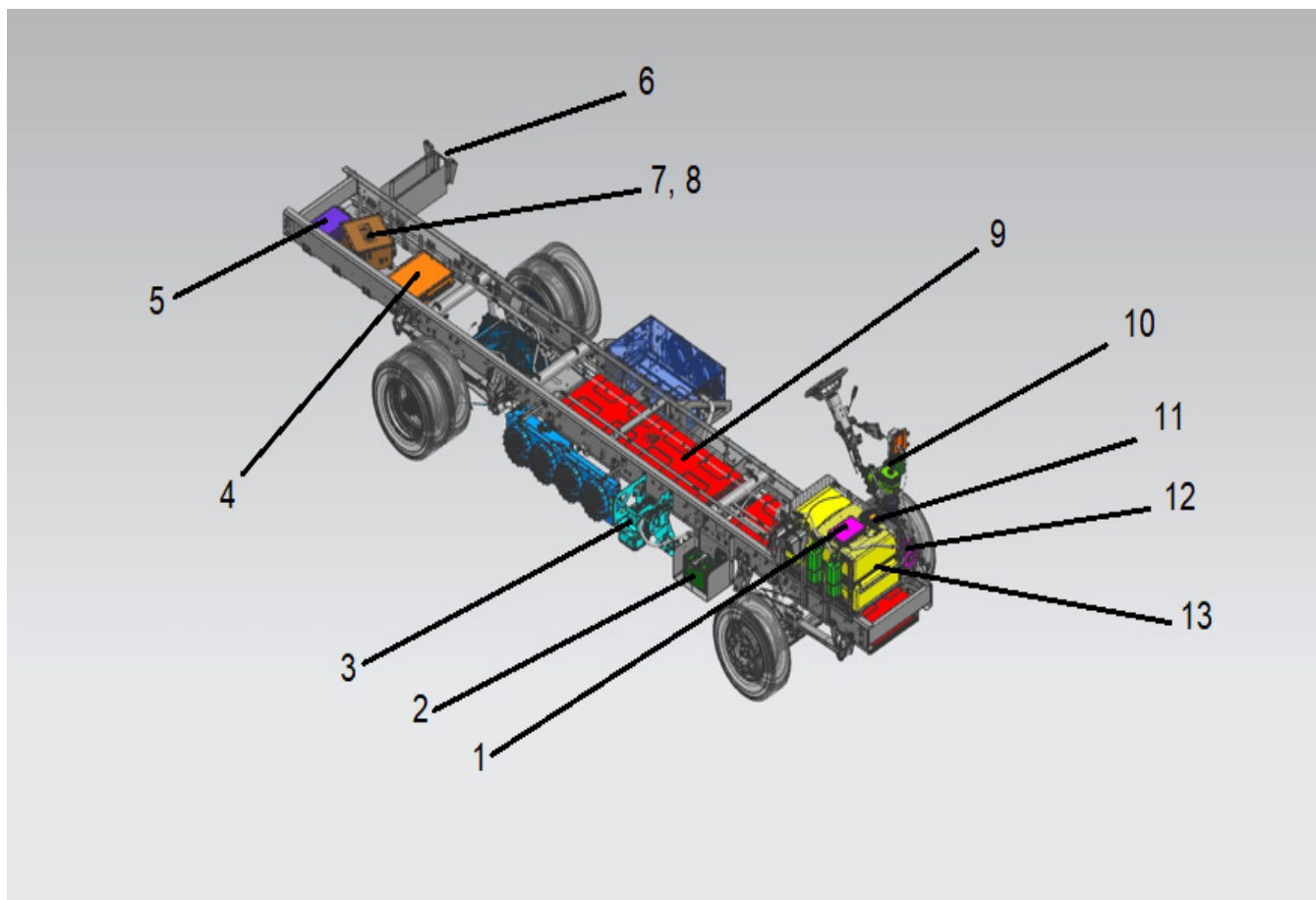
Models  
**W56**

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## EXTRACTION INFORMATION

### Component Locations



| Item | Component Name                                    |
|------|---------------------------------------------------|
| 1    | Direct Current to Direct Current (DCDC) Converter |
| 2    | 12V Battery                                       |
| 3    | Air Conditioning Compressor                       |
| 4    | High-Voltage Power Distribution Unit (HV PDU)     |
| 5    | On-Board Charger (OBC)                            |
| 6    | Charge Port                                       |
| 7    | S-Box                                             |

| Item | Component Name                  |
|------|---------------------------------|
| 8    | Manual Service Disconnect (MSD) |
| 9    | H-Pack HV Batteries (2)         |
| 10   | Brake Master Cylinder           |
| 11   | Vehicle Control Unit (VCU)      |
| 12   | Power Steering Pump             |
| 13   | C-Pack HV Batteries (2)         |
|      |                                 |

### DANGER



#### Electrocution Danger

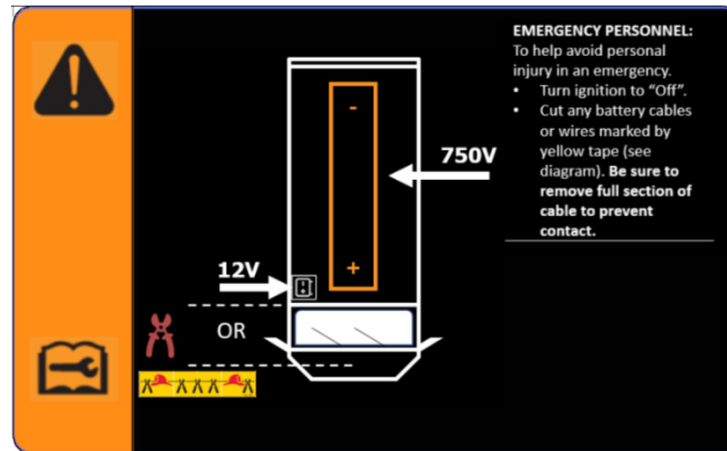
NEVER cut, breach, or touch high voltage components or cabling. Doing so could result in serious injury or death.

### WARNING

Electrical and mechanical releases may be compromised after a collision. Always wear appropriate protective equipment (Insulated Gloves, Safety Goggles).

## EMERGENCY HV CUT ZONES

This emergency cut zone label is located on two places on the vehicle. One next to the 12V battery on the passenger side and other under hood on the firewall on the drivers side. Only one of these cables needs to be cut to de-energize the HV system.



Passenger side emergency cut zone



Under hood emergency cut zone

\*After one of the emergency cables are cut, responders may still hear the BTMS and ECP fans running - The running of these fans does NOT mean the high voltage system remains active.

## 2nd responder and/or long term after accident storage HV disable:

The High-Voltage Disable and Verification procedure has three stages for disabling the vehicle.

- The following information is the first stage of HV disabling which would be used by 2nd responders and storage of damaged vehicle.

### **WARNING**

- Always perform the High-Voltage Disable and Verification procedure prior to servicing any high-voltage component or connection. Personal Protective Equipment (PPE) and proper procedures must be followed.
- Before working on any high-voltage system, wear safety glasses with appropriate side shields when within 15 meters (50 feet) of the vehicle, either indoors or outdoors.
- Cables from batteries to S-Box are still live at this time.



- Turn the ignition key on. If already on, turn the key off, wait 30 seconds, and turn back on.
- Enable service mode (see "Vehicle Service Mode" on page 6).
- Once service mode is displayed, wait 30 seconds, then turn the key off.
- Remove vehicle ignition key from the vehicle.
- Disconnect neg (-) cable from battery.



- Release first CPA (Connector Position Assurance) latch using a plastic (or non-conducting) trim tool, and move 2-stage lever to 45°.

- Dawn PPE.
- Proceed to under rear of vehicle.
- Locate MSD (Manual Service Disconnect) located at left rear behind rear axle.





7. Release second stage CPA latch with same tool.



9. Place one or more locks through the MSD latch to prevent somebody from replacing the MSD during servicing.



8. Move 2-stage lever so it stops at a 90° angle perpendicular to the MSD housing cap face, and remove MSD.



10. Start a 15-minute timer after removing the MSD to wait and let the system de-energize.

This completes the first stage high voltage disconnect procedure.



## ⚠ DANGER

The high voltage system may remain powered for up to 15 minutes after the vehicle is shut off.

## ⚠ DANGER



Lack of engine noise does not mean vehicle is off: vehicle movement capability exists until vehicle is fully shut down. Always wear appropriate protective equipment (Insulated Gloves, Safety Goggles).

## VEHICLE TOWING

Workhorse ALWAYS recommends the vehicle only be towed using a medium duty or higher flatbed tow truck.

### - Vehicle Service Mode

If vehicle is operating in a condition to do so, it is recommended to put vehicle into “Service Mode” before attempting to tow.

1. Turn the ignition key to the right, making sure not to apply brake pedal.
2. Push and hold the Workhorse button to enter Service Mode. It will display on the dash monitor.



## ⚠ CAUTION

TO PREVENT eAXLE FROM GENERATING ENERGY, DO NOT TOW VEHICLE OVER 5 MPH.

- Ensure all HV wiring is connected before attempting to roll vehicle.

### Front Tow Strap/Chain Placement

- Front Tow strap/chain should be placed around front axle beam.



### Rear Tow Strap/Chain Placement



- Rear Tow strap/chain should be placed around rear axle housing shaft.
- To ensure the steering wheel is free to rotate, always turn the ignition switch to the **ON** position when positioning the vehicle for towing or securing. This ensures the steering lock is released and all required lights are operational.
- When the vehicle is being towed, release the parking brake and ensure shift lever is placed in the Neutral **N** position.

## ⚠ WARNING

- When towing this vehicle, it is always best to assume the vehicle has power and HV exists.
- Always avoid coming in contact with the **ORANGE** cables, they are part of the HV wiring. Some wiring for the HV system runs under the vehicle.

## ⚠ CAUTION

DO NOT ATTEMPT TO JACK VEHICLE FROM CENTER REAR AXLE “PUMPKIN”. DAMAGE TO E-AXLE COULD OCCUR.

**In Case of Submersion**

Treat the W56 with the same precautions used for any submerged vehicle. Always wear the appropriate PPE. Water intrusion in damaged high voltage batteries can cause an 1) electrical discharge into the surrounding area of water, 2) ignition, or 3) explosion.